

EXERCISE 04:SQL JOINS
 NONDUMISO MAZIBUKO
 BRIGHTLEARN_DATA ANALYTICS
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QUESTION1:

```

78 | --QUESTION1:Show every user who has a subscriptions.Match users to subscriptions
79 |
80 | SELECT A.user_id,
81 |    user_name,
82 |    B.subscription_id,
83 |    start_date
84 | FROM exercise4.joins.users_table AS A
85 | INNER JOIN exercise4.joins.subscriptions_table AS B
86 | ON A.user_id=B.user_id;
87 |

```

Add parameter

Table  

	¹ ₂ ³ user_id	^A _B ^C user_name	¹ ₂ ³ subscription_id	 start_date
1	1	Nomvula	503	2026-03-10
2	2	David	502	2026-02-01
3	3	Anele	505	2026-04-05
4	1	Nomvula	501	2026-01-15

QUESTION2:

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88 | --QUESTION2:Show every subscription with it matching plan name and monthly price.
89 |
90 | SELECT A.subscription_id,
91 |    A.user_id,
92 |    plan_name,
93 |    monthly_price
94 | FROM exercise4.joins.subscriptions_table AS A
95 | INNER JOIN exercise4.joins.plans_table AS B
96 | ON A.plan id=B.plan id;

```

Add parameter

Table  

	¹ ₂ ³ subscription_id	¹ ₂ ³ user_id	^A _C ^B plan_name	¹ ₂ ³ monthly_price	
1	501	1	Basic	79	
2	504	6	Standard	129	
3	503	1	Premium	199	
4	505	3	Family	249	
5	502	2	Standard	129	

QUESTION3:

```

98  --QUESTION3:Show every viewing session that has a matching show.Include the show title and genre
99
100  SELECT A.session_id,
101         A.user_id,
102         B.show_title,
103         B.genre,
104         watch_minutes
105  FROM exercise4.joins.viewing_sessions_table AS A
106  INNER JOIN exercise4.joins.shows_table AS B
107  ON A.show_id=B.show_id;
108

```

Add parameter

Table ▼ +

	¹ ₃ session_id	¹ ₃ user_id	^A _C show_title	^A _C genre	¹ ₃ watch_minutes
1	904	7	Comedy Hour	Comedy	20
2	903	1	Crime Time	Drama	60
3	902	2	Tech Tales	Documentary	30
4	901	1	Comedy Hour	Comedy	45

QUESTION4:

```

109  --QUESTION4:Show every viewing session with the user who watched it.Only show session with matching user
110
111  SELECT B.user_name,
112         country,
113         A.session_id,
114         A.show_id,
115         watch_minutes
116  FROM exercise4.joins.viewing_sessions_table AS A
117  INNER JOIN exercise4.joins.users_table AS B
118  ON A.user_id=B.user_id;
119

```

Add parameter

Table ▼ +

	^A _C user_name	^A _C country	¹ ₃ session_id	¹ ₃ show_id	¹ ₃ watch_minutes
1	Nomvula	Johannesburg	903	702	60
2	David	Cape Town	902	703	30
3	Anele	Durban	905	705	90
4	Nomvula	Johannesburg	901	701	45

QUESTION5:

```

120 --QUESTION5:Show users alongwith their subscriptions ,the plan name, and the price.
121 SELECT A.user_name,
122        country,
123        C.plan_name,
124        monthly_price,
125        start_date
126 FROM exercise4.joins.users_table AS A
127 INNER JOIN exercise4.joins.subscriptions_table AS B |
128 ON A.user_id=B.user_id
129 INNER JOIN exercise4.joins.plans_table AS C
130 ON B.plan_id=C.plan_id;

```

Add parameter

Table ▼ +

	^A _C user_name	^A _C country	^A _C plan_name	¹ ₃ monthly_price	 start_date
1	Nomvula	Johannesburg	Premium	199	2026-03-10
2	David	Cape Town	Standard	129	2026-02-01
3	Anele	Durban	Family	249	2026-04-05
4	Nomvula	Johannesburg	Basic	79	2026-01-15

QUESTION6:

```

132 --QUESTION6:Show every user and any subscriptions they have.Users without subscription must still appear
133
134 SELECT A.user_id,
135        user_name,
136        B.subscription_id,
137        start_date
138 FROM exercise4.joins.users_table AS A
139 LEFT JOIN exercise4.joins.subscriptions_table AS B
140 ON A.user_id=B.user_id;
141

```

Add parameter

Table ▼ +

	¹ ₃ user_id	^A _C user_name	¹ ₃ subscription_id	 start_date
1	1	Nomvula	503	2026-03-10
2	2	David	502	2026-02-01
3	3	Anele	505	2026-04-05
4	4	Kabelo	null	null
5	5	Lerato	null	null
6	1	Nomvula	501	2026-01-15

QUESTION7:

```

142  --QUESTION7:Show every plan and the subscriptions on it.Plans with no subscribers must still appear.
143
144  SELECT A.plan_id,
145         plan_name,
146         subscription_id,
147         user_id
148  FROM exercise4.joins.plans_table AS A
149  LEFT JOIN exercise4.joins.subscriptions_table AS B
150  ON A.plan_id=B.plan_id;
151
152

```

Add parameter

Table ▾ +

	¹ ₃ plan_id	^A _C plan_name	¹ ₃ subscription_id	¹ ₃ user_id
1	10	Basic	501	1
2	11	Standard	504	6
3	12	Premium	503	1
4	13	Family	505	3
5	14	Mobile	null	null
6	11	Standard	502	2

QUESTION8:

```

153  --QUESTIONS8:Show every show and any viewing sessions on it.Shows that were never watched must still appear
154  SELECT A.show_id,
155         show_title,
156         B.session_id,
157         watch_minutes
158  FROM exercise4.joins.shows_table AS A
159  LEFT JOIN exercise4.joins.viewing_sessions_table AS B
160  ON A.show_id =B.show_id;
161

```

Add parameter

Table ▾ +

	¹ ₃ show_id	^A _C show_title	¹ ₃ session_id	¹ ₃ watch_minutes
1	701	Comedy Hour	904	20
2	702	Crime Time	903	60
3	703	Tech Tales	902	30
4	704	Cooking Lab	null	null
5	706	Wild Earth	null	null
6	701	Comedy Hour	901	45

QUESTION9:

```

162 --QUESTION9:Show every viewing session and the user who watched it.
163 --Sessions referencing users that do not exists must still appear (with NULL user details)
164
165 SELECT A.session_id,
166        show_id,
167        watch_minutes,
168        B.user_id,
169        user_name
170 FROM exercise4.joins.viewing_sessions_table AS A
171 LEFT JOIN exercise4.joins.users_table AS B
172 ON A.user_id=B.user_id;

```

Add parameter

Table ▼ +

	¹ ₃ session_id	¹ ₃ show_id	¹ ₃ watch_minutes	¹ ₃ user_id	^A _C user_name
1	901	701	45	1	Nomvula
2	902	703	30	2	David
3	903	702	60	1	Nomvula
4	904	701	20	null	null
5	905	705	90	3	Anele

QUESTION10:

```

175 --QUESTION10:Show every user the plan they are on(if any),and the monthly price .
176 --Users without a subcription must still appear
177 SELECT A.user_name,
178        country,
179        C.plan_name,
180        C.monthly_price
181 FROM exercise4.joins.users_table AS A
182 LEFT JOIN exercise4.joins.subscriptions_table AS B
183 ON A.user_id=B.user_id
184 LEFT JOIN exercise4.joins.plans_table AS C
185 ON B.plan_id=C.plan_id;

```

Add parameter

Table ▼ +

	^A _C user_name	^A _C country	^A _C plan_name	¹ ₃ monthly_price
1	Nomvula	Johannesburg	Premium	199
2	David	Cape Town	Standard	129
3	Anele	Durban	Family	249
4	Kabelo	Pretoria	null	null
5	Lerato	Port Elizabeth	null	null
6	Nomvula	Johannesburg	Basic	79

QUESTION11:

```

187 --QUESTION11:Show every user and every subscription ,
188 --including users without subscriptions AND subscription referencing users that do not exists
189
190 SELECT A.user_id,
191        A.user_name,
192        subscription_id,
193        start_date
194 FROM exercise4.joins.users_table AS A
195 FULL JOIN exercise4.joins.subscriptions_table AS B
196 ON A.user_id=B.user_id;

```

Add parameter

Table ▼ +

	¹ ₃ user_id	^A _C user_name	¹ ₃ subscription_id	 start_date
1	5	Lerato	null	null
2	1	Nomvula	503	2026-03-10
3	3	Anele	505	2026-04-05
4	2	David	502	2026-02-01
5	4	Kabelo	null	null
6	1	Nomvula	501	2026-01-15
7	null	null	504	2026-03-20

QUESTION12:

```

198 --QUESTION12:Subscription referencing a plan that does not exists
199
200 SELECT plans_table.plan_id,
201        plan_name,
202        subscription_id,
203        user_id
204 FROM exercise4.joins.subscriptions_table
205 FULL JOIN exercise4.joins.plans_table
206 ON subscriptions_table.plan_id=plans_table.plan_id;
207

```

Add parameter

Table ▼ +

	¹ ₃ plan_id	^A _C plan_name	¹ ₃ subscription_id	¹ ₃ user_id
1	12	Premium	503	1
2	10	Basic	501	1
3	13	Family	505	3
4	14	Mobile	null	null
5	11	Standard	504	6
6	11	Standard	502	2

QUESTION13:

```

208 --QUESTION13:Show every show and evry viewing session,
209 --including shows that were never watched AND sessions referencing shows that do not exist
210
211 SELECT shows_table.show_id,
212        show_title,
213        session_id,
214        watch_minutes
215 FROM exercise4.joins.shows_table
216 FULL JOIN exercise4.joins.viewing_sessions_table
217 ON shows_table.show_id =viewing_sessions_table.show_id;

```

Add parameter

Table  

	¹ ₃ show_id	^A _C show_title	¹ ₃ session_id	¹ ₃ watch_minutes
1	704	Cooking Lab	null	null
2	703	Tech Tales	902	30
3	701	Comedy Hour	904	20
4	702	Crime Time	903	60
5	706	Wild Earth	null	null
6	701	Comedy Hour	901	45
7	null	null	905	90

QUESTION14:

```

219 --QUESTION14:Show every user and every viewing_session,
220 -- including users with no session AND sessions referencing users who do not exists
221 SELECT users_table.user_id,
222        user_name ,
223        session_id,
224        show_id,
225        watch_minutes
226 FROM exercise4.joins.users_table
227 FULL JOIN exercise4.joins.viewing_sessions_table
228 ON users_table.user_id=viewing_sessions_table.user_id;

```

Add parameter

Table  

	¹ ₃ user_id	^A _C user_name	¹ ₃ session_id	¹ ₃ show_id	¹ ₃ watch_minutes
1	1	Nomvula	901	701	45
2	1	Nomvula	903	702	60
3	3	Anele	905	705	90
4	2	David	902	703	30
5	null	null	904	701	20
6	4	Kabelo	null	null	null
7	5	Lerato	null	null	null

QUESTION15: Show every user, every subscription, and every plan in one query-using FULL OUTER JOIN throughout.

```

232 | SELECT users_table.user_id,
233 |        user_name,
234 |        subscription_id,
235 |        plans_table.plan_id,
236 |        plan_name
237 | FROM exercise4.joins.users_table
238 | FULL JOIN exercise4.joins.subscriptions_table
239 | ON users_table.user_id=subscriptions_table.user_id
240 | FULL JOIN exercise4.joins.plans_table
241 | ON subscriptions_table.plan_id=plans_table.plan_id;

```

Add parameter

Table ▼ +

	¹ ₃ user_id	^A _C user_name	¹ ₃ subscription_id	¹ ₃ plan_id	^A _C plan_name
1	1	Nomvula	503	12	Premium
2	1	Nomvula	501	10	Basic
3	3	Anele	505	13	Family
4	5	Lerato	null	null	null
5	4	Kabelo	null	null	null
6	2	David	502	11	Standard
7	null	null	504	11	Standard
8	null	null	null	14	Mobile

BONUS QUESTIONS:

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243 | --BONUS QUESTIONS:
244 | --BONUS 01:Which users have not subscribed to any plan?
245 | --ANSWER:Kabelo and Lerato have not subscribed to any plans.
246 |
247 | --BONUS 02:Which subscriptions reference that do not exist in the users table?
248 | --ANSWER:Subscription ID:504/User ID:6.
249 |
250 | --BONUS 03:Which shows have never been watched ?
251 | --ANSWER:The Cooking Lab show and Wild Earth have never been watched.
252 |
253 | --BONUS 04:Which viewing sessions reference shows that do not exist?
254 | --ANSWER:Show ID:705/ Session ID:905
255 |
256 | --BONUS 05:Which plans have no subscribers?
257 | --ANSWER:The mobile plan has no subscribers.
258 |

```